



CNIT 27200 – Semester Group Project Spring 2025

Overall, the project will follow these development steps:

(Note: course naming standards are enforced throughout the project.)

- Design and build logical ERD (5 – 6 entities)
- Build physical ERD from the logical ERD
- Generate the DDL; confirm/modify; create and execute DDL script file
- Create and load test data
- Test the database using assigned queries
- Document the system
- Evaluate the group performance

Overall, the project is expected to follow this timeline:

- Project announced and groups assigned: **Feb 5-7 in lab sessions**
- Phase1 (design): due **Week 7, Feb/26-28, in lab sessions**
- Phase2 (DDL and Load data): due **Week 12, Apr/9-11, in lab sessions**
- Final project due (testing/SQL/final submission): **Week 14, Apr/23-25, in lab sessions**

Points of Interest

1. The semester project is worth 150 points
2. The final project, with all documentation, is **due Week 14, Apr/23-25, in lab sessions**
3. Late submissions will be penalized.
4. Each member of the group must submit a group/individual evaluation.
5. Each phase is worth 50 points for a total of 150 points.

Group Evaluations

Each group member will be evaluated (confidentially) by the other group members. Generally, all group members will receive the same score. However, an individual score may be penalized if attendance, contributed material, and the Assessment Form evaluations warrant it. If you do not turn in an evaluation on Brightspace, your own final score will be penalized.

Evaluations are due by 11:59 PM on Friday, April 25th.

Final Project Deliverables

- ☐ Cover page with Project Title, Group Number and List of Group Members
- ☐ List of Contents (basically the list of deliverables).

- ☐ Brief one page introduction/narrative description of the database (purpose, general coverage) from Phase 1 (one page), including a response to each of the following questions:
 - i. Why is the database needed?
 - ii. Who will use the database?
 - iii. What types of questions should the database be able to answer?
 - iv. What incentives will be available for those that provide data to the database?
- ☐ Table of Attendance (see below for details).
- ☐ Individual Contribution Chart (see below for details).
- ☐ PDF of final logical and physical models. **They must be readable when printed.**
- ☐ The final DDL script and the **output** from an execution of the DDL script (make sure your DDL matches the design).
- ☐ The final Data Load script and the **output** from an execution of the script.
- ☐ The test SQL (see "Testing the Database") execution and **formatted** output (sufficiently readable). The test requirements are to be divided up among the team members. For each test query, list the person responsible, and show the Oracle execution and output, numbered by SQL query number. Even though one person is responsible, the entire team will be impacted by the work (if a query is incorrect the team score is dropped). Warning: Consider some review/confirmation process to assure accuracy.
- ☐ A one page reflection from each team member about their performance and participation in the project and how these skills can be utilized beyond the classroom.
- ☐ Peer evaluation completed on Brightspace.

Table of Attendance – Submit with each phase

Any time at least 2 group members get together to discuss the project record the following:

1. Name of attendees
2. Date/time
3. Duration of meeting
4. Method of collaboration (meet in person, email, chat, Google Docs, etc)
5. Agenda for Meeting

Individual Contribution Chart and Explanation (sample) – Submit with each phase based on deliverables.

The Individual Contribution Chart identifies who worked on which tasks within the project. The Chart has group member names across the top and tasks on the left column. All group members must contribute equally for each phase in the CNIT 27200 project to receive credit.

<u>TASKS {these are just examples below – create a task list within your group}. Be as detailed as possible:</u>	Name1	Name2	Name3	Name4
Create Entities	X	X	X	X
Create Attributes		X	X	X
Create Relationships		X		X

Explanation of Contribution (each student must explain their own contribution)

Group XX

Name:
Explanation of contribution:

Name:
Explanation of contribution

etc...

Project Milestones

Project milestones will be due at various times. Each group member will be evaluated (confidentially) by the other group members. Generally, all group members will receive the same score. However, a score will be dropped one or more letter grades if the evaluations and project material warrant it due to a lack of participation.

Phase 1: Initial DB Design (50 points):

Design a normalized database (3NF) that satisfies the project requirements.

Phase 1 Submission Checklist:

- ☐ Include a cover page with database name, group members, date and phase #.
- ☐ Submit a one page description of the database, including a response to each of the following questions:
 - i. Why is the database needed?
 - ii. Who will use the database?
 - iii. What types of questions should the database be able to answer?
 - iv. What incentives will be available for those that provide data to the database?
- ☐ Create a list of entities (tables) and attributes (columns). **The project requires 5-6 entities and more than 3 attributes per entity.**
- ☐ Provide some example data (records) that might be stored in your database.
- ☐ Table of Attendance
- ☐ Contribution Chart and Explanation (describe in detail how each member of the group contributed to this phase of the project). All group members must participate for credit.
- ☐ Recap of meeting with your TA to discuss progress/questions/help with Phase 1.
- ☐ Create a PDF, print it out, and submit it to the TA in your lab session with the the following two items:
 - ☐ An Entity Relationship Diagram **logical model** including:
 - i. Entities (5-6 entities)
 - ii. Attributes

- a. each entity must have a minimum of 3 non-primary key attributes, plus the primary key attribute(s)
- iii. Primary Keys
- iv. Foreign Keys
- v. Identifying and Non-Identifying Relationships
- vi. Parent-child verb phrases
- vii. Text box with title, date, and group information
- viii. Other features as instructed in the documentation
- ☐ An Entity Relationship Diagram **physical model** that displays appropriate data types and sizes:
 - i. Text box with title, date, and group information
 - ii. Verify that all datatypes and sizes are visible on the submission.

NOTE: The paper of your logical model and physical model need to be readable. Before submitting the printed papers to your TA, test the readability of the output.

Advice:

- 1) As a group, determine a method for keeping your files accessible to all members. You can use Google Drive, OneDrive, Teams, etc.
- 2) Verify that all of your entities have primary keys
- 3) You should include some attributes in your database design that can be calculated later when testing the database (NUMBER datatype; Price, Salary, Fee, etc)
- 4) Review each entity and verify that each non-keyed attribute depends on its primary key (remember - 2nd Normal Form).
- 5) Review each entity to verify that there are no transitive dependencies (remember – 3rd Normal Form).
- 6) Do not create entities with only a few attributes. Review the entity and reflect on why it is needed. Maybe there are more attributes that can be populated and stored in the database. Be creative. Google the topic of your database to generate ideas.
- 7) Utilize associative entities (no many to many relationships).
- 8) Verify that every item in your model is visible in the submission. If datatypes are hidden, or can only be partially seen, points will be deducted.
- 9) If a group member does not contribute to the project, and there is no detailed description of their participation in the submission, that group member will get a zero for the phase.